USED CAR
PRICE ANALYSIS

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INTRODUCTION

The used car market in India has grown rapidly, offering an affordable alternative for consumers looking for high-quality vehicles at lower prices. Platforms like Cars24 provide a comprehensive listing of used cars, including various car companies, fuel types, and transmission systems. This project analyzes the variation in car prices across different factors to help consumers make better decisions when purchasing used cars.

However, the price of used cars can vary significantly based on numerous factors such as:

- Company Name
- Fuel Type
- Distance Driven
- Prices

PROBLEM STATEMENT

Understanding the factors that influence the price of used cars is critical for both buyers and sellers. This project seeks to answer the following:

- How do car prices vary based on company, fuel type and Distance Driven?
- What are the best value cars in terms of price and quality?
- What is the impact of car age and distance driven on the price of used cars?

OBJECTIVES

The primary objectives of this project are:

- To analyse car price variation across different companies in the used car market.
- To identify cars that provide good quality at a lower price (value-for-money cars).
- To study the impact of factors such as car age, distance driven, fuel type and transmission on car prices.
- To study the company-wise car availability in each city.

METHODOLOGY

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and visualization. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

1. **Data Collection:**

   The first step of the methodology involves gathering data on used cars. For this project, data is collected using web scraping from the online platform Cars24, a popular marketplace for buying and selling used cars.

- **Tools Used:** Python libraries such as Selenium, BeautifulSoup, and Pandas are utilized for web scraping and data handling.
- **Process:**
  - Selenium is used to automate the process of visiting the Cars24 website and navigating through the car listings.
  - Once the data is loaded, BeautifulSoup parses the HTML content and extracts relevant data fields such as Car name and model, Car Company, Fuel type (Petrol, Diesel, CNG), Distance Driven (Distance covered), Price (in lakhs).
  - The extracted data is then stored in structured formats (like Pandas Data Frame) for further analysis.

2. Data Cleaning:

After collecting the raw data, the next step is Data cleaning. The goal of this phase is to ensure the dataset is free from errors, missing values, and irrelevant information.

Cleaning Steps:

- Removing the Duplicates and the Null Values.
- Replacing the missing values with relevant placeholders.
- Converting the Datatypes of the Columns.
- Run the basic EDA (Exploratory Data Analysis) using python.

3. Data Normalization:

Normalizing prices and other numeric fields to standard formats. For example, prices are converted to a uniform format (in lakhs), and also Distance Driven values are converted into the numerical fields.

4. Data Visualization:

After Cleaning the dataset, the next step is Data visualization, which helps in understanding the relationships between various factors like car company, price, Distance Driven and Fuel type.

Tools Used:

- **Power BI:**

  For creating interactive and dynamic dashboards that allow users to filter data and explore trends across different car companies, fuel types, and etc.

- **Creating Visualizations:**

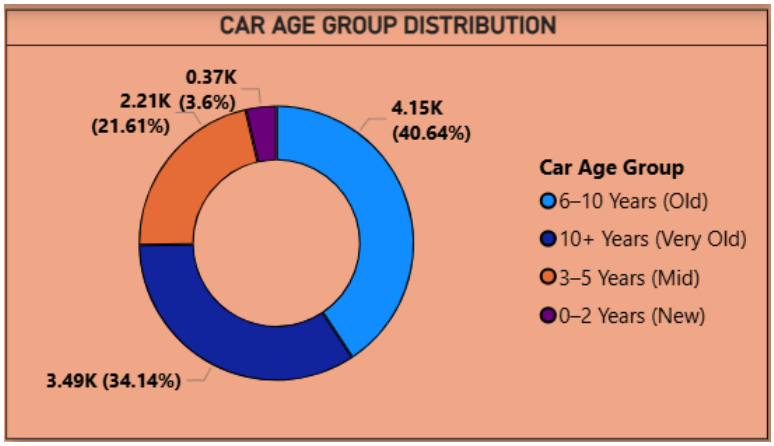
  - No of Used Cars by Company
  - Car Price Variation by Company
  - Used Cars Distribution by Fuel Type
  - Car Price Segment Distribution
  - Price VS Distance Driven Distribution
  - Value For Money Cars Comparison Across Cities

- **Creating Dashboard:**

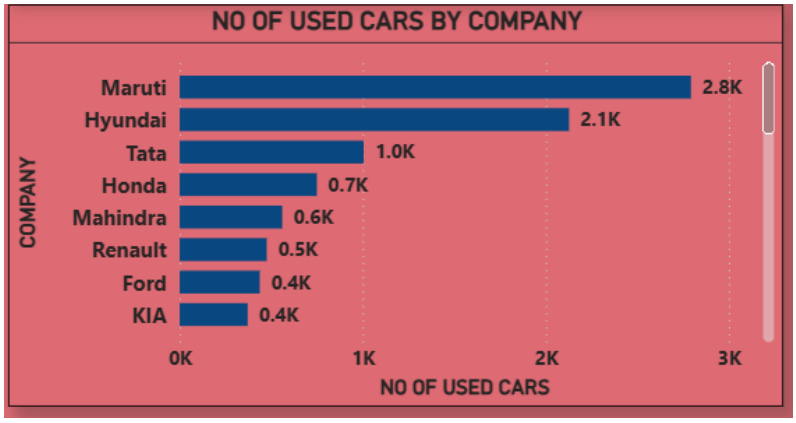
  - Once the visualizations are created, you can combine them into a comprehensive dashboard.
  - Users can filter the data by Company, City and fuel type to observe the price variations across different categories.

RESULTS

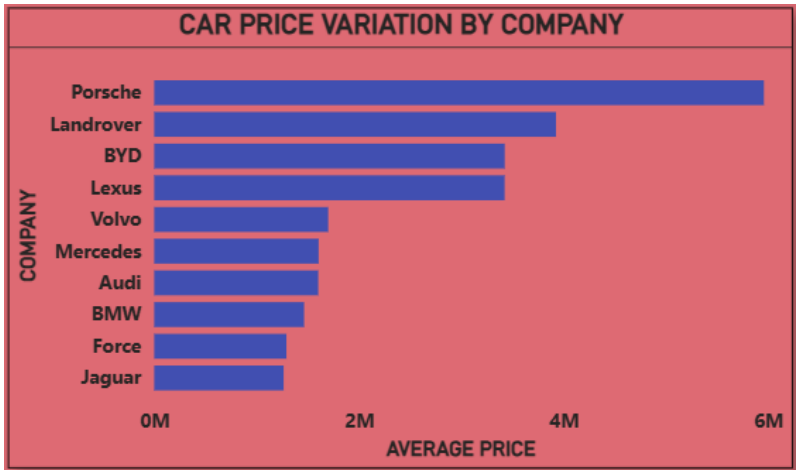
- Car Age Group Distribution:



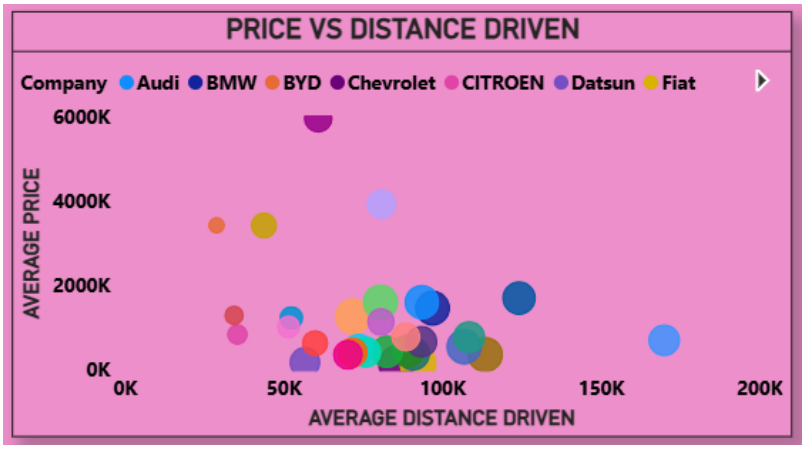
- No Of Used Cars by Company:



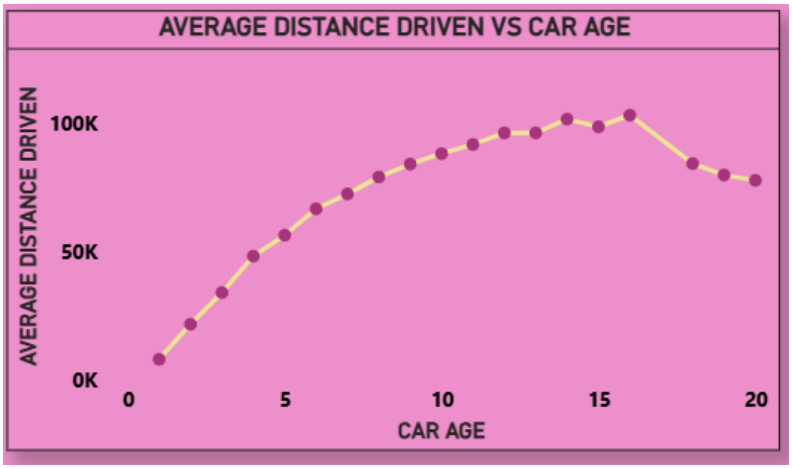
- Car Price Variation by Company:



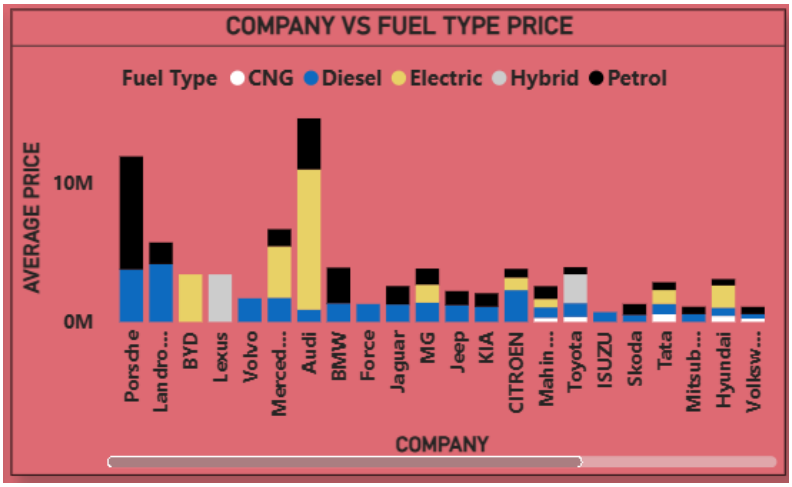
- Price Vs Distance Driven:



- Average Distance Driven Vs Car Age:



- Company Vs Fuel Type Price:



KEY FINDINGS

- **Car Price Variation by Company:**

  Porsche has the highest average car price (around ₹6M), mass market companies such as Maruti, Tata, and Hyundai fall into the lower average price range, making them more affordable.

- **No. of Used Cars by Company:**

  Maruti dominates the used car market with approximately 2.8K cars, other companies like Honda, Mahindra, Renault, Ford, and KIA have comparatively lower representation.

- **Average Price vs Car Age:**

  Average car price decreases consistently as car age increases, confirming depreciation over time. The steepest price drop is observed between 2 to 8 years of car age.

- **Company vs Fuel Type Price:**

  Premium companies show higher average prices across Petrol, Diesel, and Electric fuel types**.** Electric and Hybrid vehicles are mostly concentrated among high end companies**,** contributing to their higher average prices.

CHALLENGES FACED

- Initially, Data extraction using BeautifulSoup was unsuccessful due to the dynamic nature of the website and JavaScript-rendered content. To overcome this limitation, Selenium was used to automate browser interactions and successfully fetch the required data.
- The dataset contained missing values, inconsistent data formats (such as price, distance driven, and EMI), and incorrect or unrealistic entries. These issues were overcome by performing data cleaning techniques including handling missing values, standardizing units and formats, correcting inconsistencies, and removing or treating invalid outliers to ensure data accuracy and reliability.

```
[13]: df.head(5)
```

[13]:

	Car Name	Company	Year of Manufacture	Distance Driven	Fuel Type	Transmission Type	EMI	Price	Car Age	City
0	Maruti Wagon R 1.0	Maruti	2018	40,285 km	Petrol	Manual	EMI ₹5,896/m*	₹3.34 lakh	8	Ahmedabad
1	Maruti Alto	Maruti	2019	38,068 km	Petrol	Manual	EMI ₹3,959/m*	₹2.24 lakh	7	Ahmedabad
2	Maruti Swift	Maruti	2019	40,477 km	Petrol	Manual	EMI ₹7,890/m*	₹4.47 lakh	7	Ahmedabad
3	Hyundai Grand i10	Hyundai	2018	62,098 km	CNG	Manual	EMI ₹9,939/m*	₹4.47 lakh	8	Ahmedabad
4	Hyundai Elite i20	Hyundai	2017	75,093 km	Petrol	Manual	EMI ₹8,371/m*	₹4.28 lakh	9	Ahmedabad

```
[23]: df.head(5)
```

[23]:

	Car Name	Company	Year of Manufacture	Distance Driven	Fuel Type	Transmission Type	EMI	Price	Car Age	City
0	Maruti Wagon R 1.0	Maruti	2018	40285	Petrol	Manual	5896	334000	8	Ahmedabad
1	Maruti Alto	Maruti	2019	38068	Petrol	Manual	3959	224000	7	Ahmedabad
2	Maruti Swift	Maruti	2019	40477	Petrol	Manual	7890	447000	7	Ahmedabad
3	Hyundai Grand i10	Hyundai	2018	62098	CNG	Manual	9939	447000	8	Ahmedabad
4	Hyundai Elite i20	Hyundai	2017	75093	Petrol	Manual	8371	428000	9	Ahmedabad

CONCLUSION

This analysis shows that car price is mainly influenced by company, vehicle age, and usage, with clear depreciation as car age and distance driven increase. Petrol cars dominate the market and most used cars fall in the mid-price and mid-age segments. By understanding factors such as company, car age, usage, and customer demand, sellers can price vehicles more accurately. These insights help improve sales planning, meet customer expectations, and stay competitive in the used car market.

REFERENCES

- **Data Sources:** Cars24 Website
- **Tools Used:** Python (Selenium, Beautiful Soup, Jupyter Notebook), Power BI.